Journal of Computer-Mediated Communication, 2023, **29:1**, zmad042
https://doi.org/10.1093/jcmc/zmad042
Full-length Research Article

Codesigning community networking literacies with rural/remote Northern Indigenous communities in Northwest Territories, Canada

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Abstract

Digital literacy research and practice typically presume certain conditions, such as an urban orientation and adequate, affordable access to connectivity and devices. But these conditions are not universal; for example, people in small, rural/remote Indigenous communities may seek to balance connectivity challenges and digital innovations with land-based living specific to place and community. Drawing on efforts to broaden critical digital literacies to support Indigenous sovereignty, we consider how overlapping contexts of places, communities, and infrastructures intersect in the cocreation of appropriate digital literacy. Specifically, we discuss a series of virtually facilitated, participatory workshops that utilize "hacker literacies" and "infrastructure literacy" to reimagine connectivity infrastructure and demonstrate the potential of community networking in, with, and by rural/remote Indigenous communities. We also reflect on limitations of this work and identify lessons for future projects.

Lay Summary

Persistent digital divides exist between Northern and Southern Canada, and between rural/remote and central communities in Northern Canada. Along with large-scale commercial telecommunications projects, locally designed and deployed networks offer potential solutions to improve connectivity. Community Networks involve residents in the deployment, ownership, and control of digital infrastructures. However, past research has determined challenges in community efforts to deploy and sustain local networks. In this study, we document the process we used to codesign locally relevant community networking resources—an orientation made more salient given the context of COVID-19 and the desires of rural/remote Indigenous and Northern communities for ownership and control over digital infrastructures. We reflect on the limitations of this approach and suggest lessons learned for future projects.

Keywords: digital inclusion, digital literacy, digital divide, community networks, Northern Canada, Indigenous Peoples.

Digital Literacy research and practice has historically presumed certain conditions, such as adequate, affordable access to broadband connectivity and devices, and a desire to learn digital applications developed in and for urban settings (Gray et al., 2018; Sikkema et al., 2019). But these conditions are not universal; for example, people in small, diverse, and rural/remote Indigenous communities located in Denendeh and Inuvialuit Nunangat—or the Northwest Territories (NWT), Canada—may choose to balance connectivity challenges and digital innovations with land-based living specific to place and community. Drawing on efforts to broaden critical digital literacies to support Indigenous sovereignty (Duarte, 2017; Gaertner, 2023; Wemigwans, 2018), we consider how overlapping contexts of places, communities, and infrastructures intersect in the cocreation of appropriate digital literacy. Specifically, we discuss a series of virtually facilitated, participatory workshops that utilize "hacker literacies" (Santo, 2013) and "infrastructure literacy" (Parks, 2010) to reimagine connectivity infrastructure and demonstrate the potential of community networking in, with, and by rural/remote Indigenous communities. We present our efforts to date, situate our conceptual framework and codesign methodology, introduce our mesh networking kits, and consider how our findings can support more localized approaches to community networking and infrastructure literacy. We also identify the limitations of this work, reflecting on lessons learned for future projects.

As an international team of researchers, technologists, and community members who are both non-Indigenous and Indigenous, we recognize the challenges of collaborating from very different geographic and epistemological positions (Darnell, 2018; Stiegman & Castleden, 2015). Our approach to digital literacy theory and practice is informed by critical pedagogy (Freire, 1970; Kumashiro, 2000) and place-based and contextual literacies (Sikkema et al., 2019). Our critical stance attends to unequal power dynamics, intersectional and overlapping forms of inequity, and foregrounds the social practices involved in the development and operations of socio-technical systems (Carmi & Yates, 2020; Nichols & Stornaiulo, 2019; Pangrazio, 2016; van Deursen & Helsper, 2015). Yet our project also surfaced structural and practical limitations of this work; we document these issues with the goal of supporting future research and practice.

Digital literacy encompasses multiple topics, from learning to use devices, to coding software, to determining sources of misinformation (Bawden, 2008; Gilster, 1997). These skills-

Associate Editor: Teresa Correa
Received: 2 May 2023. **Revised:** 12 August 2023. **Accepted:** 21 September 2023
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oriented initiatives typically assume learners have adequate access to digital technologies, such as the network infrastructures providing connectivity (Sharpe et al., 2010). As a result, digital literacy can be alienating to individuals or groups unable or reluctant to adopt digital technologies (Secker, 2018). While there is an extensive body of literature on digital literacy skills development in well-connected formal settings such as schools, universities, and libraries, here we focus on a lesser developed space—contexts where access to digital connectivity and associated technologies is limited, unreliable, and/or expensive. This orients our work toward *infrastructural literacies*; that is, how the material aspects of digital networks are planned, built, managed, and understood by end users (Forsler, 2018; Gray et al., 2018; Greig, 2018; Lithgow et al., 2022). As Parks (2010) writes, "There is a tendency to overlook the uniqueness of particular nodes in a network: their physical form, the stories of their development, or the practices that surround them once they are activated" (p. 345; see also Lithgow, 2021; Mattern, 2016).

Through a series of participatory online workshops and semi-structured interviews, we examined and documented these aspects of infrastructural literacies. The Research Question guiding this work was: *In what ways can the social and technical aspects of community mesh networking be adapted by people in NWT communities to develop appropriate forms of infrastructural literacy?* Our iterative codesign research process asked project participants about their boundaries, goals, interests, and suggestions. Along with guiding the creation of resources for community networking literacies, their responses informed the material development of open-source, portable and offline-first mesh network kits. They also surfaced several limitations inherent to this work, including for urban researchers attempting to work collaboratively with rural/remote Indigenous communities in Northern Canada.

Situating digital connectivity in the NWT, Canada

Recent digital inclusion literature reflects a shift from technically oriented digital divides to a broader focus on digital inequalities, including in relationships between the deployment and operations of digital connectivity and "offline" inequities (Robinson et al., 2020). This article is situated in relation to geographic digital divides tied to settler colonialism and extractivism in Indigenous territories in the Western Arctic regions of Canada (Coulthard, 2014). These lands are ancestrally referred to as Denendeh, named after the giant who fell from the stars. The boundaries imposed through the administrative formation of the NWT (formerly referred to colonially as Rupert's Land) cut and divide Denendeh and Inuit Nunangat. Over time, settled municipalities replaced ancestral land-based, migratory communities, in part because of mandated requirements in the *Indian Act* for Indigenous youth to attend residential school, as well as the emergence of European/settler colonial institutions such as government offices, church missions, resource extraction, and trade (such as through the Hudson's Bay company), and the Northwest Mounted Police.

Today the three Northern territories (Yukon, NWT, and Nunavut) comprise a total area of 1,731,408 square kilometers (39.52% of the land in Canada) but only one-third of one percent (0.33%) of the population. As of Q2 2023, their population sizes are: 44,412 in the Yukon, 45,668 in the NWT, and 40,715 in Nunavut (Statistics Canada, 2023). Outside of territorial capitals and regional hubs, people live in small, widely dispersed rural communities with a median population of 561. Cold temperatures limit the construction season, and many communities rely on boats (in summer), ice roads (in winter), or airplane transportation. Notably, 12 out of the 33 NWT communities can only be accessed by regional small airlines (Government of NWT [GNWT], n.d.a, n.d.b). The specific communities participating in this project range in size from 205 to 628 residents and have limited transportation links (two are fly-in communities, one is accessible by ice road in winter, and another is accessible by boat or ice road) (GNWT, 2022). Electrical networks are similarly underdeveloped, with most communities reliant on diesel-fired power plants. In summer of 2023, intensive forest fires ripped through the territories and near many communities, highlighting their susceptibility to the cataclysmic impacts of climate change.

Despite a history of community-led innovation, Northerners experience similar inequities in communication infrastructure. For example, in the early 1970s a system of high-frequency two-way "bush radios" connected local nursing stations and Hudson Bay stores, and supported communication between communities and during land-based activities such as fishing, hunting, and trapping (Hudson, 1974). When regulatory obligations required telecommunications companies to provide analog telephone services in the late 1970s and early 1980s, smaller communities had to settle for a single pay phone, along with outdated equipment and periodic maintenance undertaken by fly-in technicians. Partly due to the lobbying efforts of Indigenous organizations, residential telecommunications services became more widespread in the 1980s. However, the pattern continued in the 1990s, when inadequate telecommunications infrastructure precluded community access to digital applications like email and the Internet (McMahon et al., 2017).

Today, despite years of advocacy by Indigenous and public interest groups, the major commercial telecom provider (operating as a regulated monopoly) continues to deliver expensive, unreliable connectivity far below the Basic Service Objective—defined by the Canadian Radio-television and Telecommunications Commission as speeds of 50 Mbps download/10 Mbps upload (Canadian Radio-television and Telecommunications Commission [CRTC], 2016). Publicly available data on connectivity infrastructure and services in rural/remote NWT communities remains limited but indicates that most communities experience slow, unreliable connectivity and expensive data caps, among other challenges (DigitalNWT, n.d.; McMahon & Akçayir, 2022). Until very recently, most Northern communities had DSL or cable access and, to a lesser extent, satellite Internet with data caps. Fibre connections are available only in a few larger populated communities, such as the capital of Yellowknife. Until the recent uptake of low-earth orbital satellite connections, small communities remained dependent on a single incumbent service provider, resulting in higher fees for Internet services that were both slow and subject to monthly limitations.

At the same time, the diverse Indigenous Peoples living in these areas have led successful local and regional community development initiatives since time immemorial—demonstrated contemporarily through vibrant local arts groups, community radio stations, and communal freezers. These

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initiatives include connectivity projects ranging from Jean Marie River First Nation's fixed wireless network to complex regional organizations such as K-Net services in Northern Ontario and Tamaani Internet in Nunavik (Carpenter, 2010; First Mile Connectivity Consortium [FMCC], 2018; Roth, 2013; Savard, 1998). These projects demonstrate Indigenous deployment of and sovereignty over network infrastructure and services in areas with a limited case for private sector telecommunications investment.

In the NWT, several recent projects have been developed or announced to improve connectivity for communities (CRTC, n.d.; Dempster Fibre Project [DFP], 2022; Mackenzie Valley Fibre Link [MVFL], 2021). Along with benefits, these projects may also bring challenges such as expensive services and an influx of inappropriate and culturally irrelevant content. They may also limit local economic benefits, as revenues are extracted from communities rather than being invested and circulated locally. For these reasons it is important to learn from Northern residents how best to mitigate risks and utilize the opportunities of enhanced connectivity. Our project involves learning together how to shape educational resources about digital infrastructures that may enable locally managed and self-determined pathways of digital development. We draw from the global community networking movement, which presents ways to facilitate local codevelopment, management, ownership, and sustainability of community network (CN) infrastructures.

Community mesh networks in rural/remote Northern Indigenous contexts

CNs involve communities in the deployment, ownership, and/or control of digital infrastructures (Antoniadis, 2016; Baca-Feldman, 2021; Belli, 2017). CNs often emerge from necessity, as individuals and groups excluded from or beholden to commercial telecommunications networks must build and operate their own connectivity solutions. There is a large social component to this work: CN projects can coalesce a community around shared desires and experiences, as well as frustrations around barriers to Internet affordability and accessibility. Local wireless mesh networks include early work undertaken by the BC Wireless Network Society and Île Sans Fil ("Wireless Island") in Canada (Powell & Shade, 2006), and the Commotion Wireless Project in the USA (Reynolds et al., 2011). Recent projects include efforts in cities including Detroit (Greig, 2018), New York (NYCMesh), Toronto (ToMesh), and Cape Town (V-NET), rural areas in Mexico (Telecomunicaciones Indígenas Comunitarias), Argentina (Altermundi), South Africa (Zenzeleni), Catalonia (GuifiNet), and elsewhere (e.g., Freifunk in Germany).

These efforts include a history of Indigenous-led initiatives that position CNs as a means for Indigenous nations to exercise autonomy and sovereignty (Duarte, 2017; Makaawaawa, 2019; McMahon et al., 2011). Such work resists the individualizing tendencies of technology development to re-focus on collective processes (Gurstein, 2012; Reina-Rozo, 2019). Issues negotiated through these projects include intellectual property, language control and revitalization, content production, community memory, and protection of Ancestral Knowledge, as well as critical perspectives on digital forms of capitalism and colonization (Gaertner, 2023). Such efforts reorient technical innovation beyond discourses and processes tied to modernization and technology transfer toward calls from Global South scholars for culture-centered (Dutta, 2020) and territory-centered (Hinojosa & Baca-Feldman, 2021) projects rooted in local and community contexts.

The work undertaken by CN projects is enabled by recent technical and policy developments and objectives: open access to fiber backhaul, availability of unlicensed spectrum, low-cost networking equipment, and open-source hardware and software. As Song et al. (2018) write: "The technologies available are now being exploited in many imaginative forms by communities around the world to meet their communication needs" (p. 9). More inclusive and intuitive technical designs are lowering the threshold of expertise required to deploy and manage CNs. This includes decentralized design and manufacturing of network components, use of local materials and expertise, and repurposing/upcycling of old network hardware (Tecnología De Raíz, 2022).

Despite this potential, Song et al. (2018) point out barriers to the widespread adoption of wireless mesh CNs, including a lack of awareness of the opportunity and relative ease of deployment, the current state of enabling policy and regulation (including access to wireless spectrum), limited funding for capital expenses, limited technical expertise, and governance issues of ownership and control of projects, services, and applications. These and related observations (e.g., Greig, 2018; Lithgow et al., 2022) raise important questions about the inclusive design and sustainability of CN projects (Kende, 2016). Although CNs can leverage a range of technologies, Srivastava (2019) underscores that to be sustainable, they must be contextually relevant to Indigenous communities. Taking inspiration from groups like REDES A.C., Rhizomatica, APC, and others (for an extensive list of CN projects see APC (2022)), our project works alongside community members to shape digital literacy resources to support pathways toward sustainable, community-led design, adoption and diffusion of digital connectivity.

Exploring community networking literacies through DigitalNWT

The collaborative codesign of community networking literacies described in this article is embedded within DigitalNWT, a participatory action research project that worked with community members across the NWT to research and deliver digital literacy training (DigitalNWT, n.d.). DigitalNWT germinated after a two-year project with the Gwich'in Tribal Council (2017–2018), where researchers from the University of Alberta traveled to Dinjii Zhuh (Gwich'in) communities to learn about community innovation and digital literacy (McMahon et al., 2018). Discussions on subjects including local connectivity and digital content were generally well received and underscored the importance of hands-on and locally oriented learning resources and activities.

From 2019 to 2022, the DigitalNWT team grew to include researchers and technologists in Mexico and Czechia, and community members from both non-Indigenous and Indigenous backgrounds. DigitalNWT is led by a Steering Committee of four Indigenous organizations: Gwich'in Tribal Council, Inuvialuit Regional Corporation, Sahtú Renewable Resources Board, and Tłı̨chǫ Government. These organizations guided the work of NWT-based educational organizations, including the Smart Communities Society (NWT) and Aurora College (an NWT post-secondary institution), along with the University of Alberta. This involved codesigning

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three introductory digital literacy courses that adult educators from Aurora College and Smart Communities Society delivered free of charge. Content covered basic Internet concepts such as digital content, devices, and social media, as well as CN mapping workshops, exercises, and outcomes (DigitalNWT/McNally et al., 2022).[1] Course 2 focuses on wired and wireless broadband technologies and community engagement—concepts made tangible through a guided learning exercise, "Make the NetWork," whereby students collaboratively build a tabletop CN model from a local map and 3D-printed pieces. The completed model (see Figure 1) is used to generate discussion regarding how, why, where, and to what ends a community is connected.

Following the "Make the NetWork" activity, the DigitalNWT team began planning a course to further understanding of local ownership and connectivity infrastructure through the deployment of a small-scale demonstration CN. Given limited connectivity in rural NWT communities, the team envisioned a locally oriented, "offline-first" system that could operate independent of the wider Internet. Course participants would build and test a small-scale (two-to-three node) wireless mesh network, which involves sharing local connectivity and content via unlicensed spectrum through a decentralized mesh topology. This network configuration pushes control of networks and content to edge devices (such as servers storing local content) and allows for locally determined deployment, an organic approach Duarte et al. (2021) describe as emerging "as a squash plant grows." Local access points (APs) enable the expansion of the mesh network and provide local access to open source applications that enable video and voice calling, network monitoring and management, web browsing, and media streaming. Such a system is oriented toward providing local connectivity within a fixed geographic location—it does not include the backhaul transport required to connect communities with one another. However, the units can provide such regional connectivity if connected to a transport link such as a satellite terminal, fiber optic link, or wireless infrastructure. In this context, the kits would ultimately serve two purposes. First, to anchor a course about community networking by giving participants the chance to learn about CNs and deploy a small scale one in practice. Second, to provide a means for the team to deliver other digitized instructional content in communities that otherwise lacked adequate connectivity.



Figure 1. Make the "NetWork" Guided Learning Activity in Tuktoyaktuk, NT (April 2021). Photo credit: Daniel Dokunmu.

To facilitate these activities, DigitalNWT partnered with Wakoma, a social enterprise focused on localized, open-source connectivity solutions. The team planned a customized mesh networking kit combining open hardware designs with offline-first open-source services. The kit consists of openly licensed 3D-printed components and network hardware (a 10th generation Intel NUC server, a Netgate SG-1100 firewall, a Ubiquiti Flex switch, and three Ubiquiti mesh APs), a patch panel to simplify wiring, and shelving. In addition, the kits included a multi-port USB hub and a 20,000 mah power bank which could serve as an uninterruptible power supply. The system operates on license-exempt 2.5 GHz and 5 GHz spectrum bands and supports common Wi-Fi standards, which enables users to connect to each wireless mesh AP at a range of 140 square meters. This hardware is housed in an easily transportable carrying case. Despite its small size, each kit can connect hundreds of simultaneous users to each other and provide local content and services regardless of Internet access, at a cost of approximately $3,000 CAD. The kits include operating software that allows for both autonomy in use and the ability to send and receive backups, content, updates, and support between global and local servers. Furthermore, the design allows for highly distributed manufacturing through a hardware guide, wiring diagram, and videos illustrating assembly steps.

Codeveloping both the mesh networking kits and the associated digital literacy course involved a series of engagements and reflections with NWT residents. As we prepared the kits, it was important to engage Northerners in design choices. In the next section, we discuss the interviews and workshops used to inform our community networking literacy codesign process.

Methodology: preparing the codesign process

Informed by community-based participatory action research (Beaton et al., 2017; Essel et al., 2021; Hinojosa & Baca-Feldman, 2021), we adapted commonly practiced codesign steps: stakeholder consultation (re-framed as identifying community desires, interests and concerns), problem definition, idea generation, concept testing, prototype testing, product manufacturing, and evaluation (Arsenault et al., 2019; Bradford et al., 2018; Duarte et al., 2021). These steps reflect iterative cycles of discussion and development that build upon one another in response to feedback from end users (Quarry & Ramirez, 2009). This work is also inspired by recent efforts to use workshops as boundary objects that bridge parallel goals of infrastructure literacy and design (Peer & DiSalvo, 2019). The social distancing requirements imposed by the COVID-19 pandemic shaped and constrained these activities, while also necessitating innovations in "pandemic pedagogies" (Smythe et al., 2021). While we recognize the importance of community-oriented, in-person, land-based research and education in Northern Indigenous contexts (e.g., Ballantyne, 2014), the circumstances of our project required us to meet via remote, digitally enabled connections. Despite limited Internet access in many Northern households, we connected with participants at community anchor institutions with adequate connections, such as adult learning centers.

As summarized in Table 1, our process involved five codesign workshops and two rounds of semi-structured

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Table 1. Summary of project activities

Date	Activity
February 2021	Recruitment workshop
March 2021	Intake interviews
May 2021	Codesign Workshop 1: Let's Build the Black Box
June 2021	Codesign Workshop 2: Customizing the Interface and Picking Applications
Summer 2021—Spring 2022	Codesign Workshop 3/Paired Support Sessions: Installing and Configuring Mesh Networking Kits
October 2021	Codesign Workshop 4: Let's Test Out the Apps
February 2022	Codesign Workshop 5: Administration and Governance of Mesh Networking Kits
March 2022	Exit interviews

interviews. Participants included a diversity of NWT residents with respect to location, size of community (urban/rural), age, gender, ethnic background (representative of Indigenous Nations and non-Indigenous Peoples), existing level of technical expertise, and profession. While involvement fluctuated over time, the number of participants ranged from 18 individuals (Intake interviews) to seven individuals (Exit interviews). Formal interviews lasted 60–90 minutes and were conducted via videoconference or in one case telephone (due to limited bandwidth). Online workshops lasted approximately 3 hours. Throughout, NWT participants reflected on their interests, goals, and desires for the project, while providing feedback about limitations and shortcomings.

To analyze the interviews and workshop recordings, we used transcripts generated by the Zoom auto-transcription feature and used inductive content analysis (Saldaña, 2021; Yin, 2011). This involved coding transcripts, identifying themes, and arranging coded responses according to themes (Emerson et al., 2011). We noted points of agreement/disagreement, and the research team discussed and resolved them.

The project began in February 2021 with a recruitment workshop involving Northern residents with expertise in technical, educational, and research fields. Most of these individuals had an existing personal and professional relationship with the project team. An issue made apparent in the first workshop was the large number of participants, which limited opportunities for substantial feedback and discussion. Some participants experienced difficulties joining due to limited bandwidth.

Following the workshop, a University of Alberta researcher conducted one-on-one interviews focused on two short presentations based on the curriculum plan described above. Conceptual aspects of the course focus on how to communicate about CNs to a non-technical audience while considering the diverse cultural contexts of NWT communities. Technical aspects considered mesh networks that incorporate open-source technologies designed for local content and offline applications. Interview questions covered a variety of themes, including who to include in the codesign process, initial impressions of the mesh networking equipment, understanding the opportunities, advantages, and challenges of a locally owned CN, and learning how participants want to collaborate moving forward. Intake interview findings are discussed in detail below.

Based on interest expressed in the initial workshop and interviews, we recruited NWT-based innovation hubs. Drawing from literature on "technology stewards" (e.g., Gow et al., 2020; Wenger et al., 2009), "social animators" (Crocker, 2008, p. 71), "infomediaries" (Hudson, 2013), and "ICT4D champions" (Renken & Heeks, 2013), we envisioned the NWT hubs as a resource to provide feedback on design choices, document learning processes (e.g., videos and photos of building kits), review and propose course content, and help promote the project. Hubs consisted of three to five people from four NWT communities—Yellowknife, Inuvik, Déline (now based in Norman Wells), and Whatì—as well as individuals from Tsiigehtchic and Fort Good Hope. These innovation hubs represent a diversity of geographic locations, population sizes, and connectivity levels (see Figure 2). Participants ranged from highly skilled network technicians and post-secondary educators to people new to community networking.

Between May 2021 and May 2022, these NWT innovation hubs worked through five remotely delivered, online (Zoom) codesign workshops that covered different codesign themes (discussed in detail below). During semi-regular meetings we assembled, configured, and tested hardware and software for the mesh networking kits. These workshop discussions directly informed the material and software design of the kits.

Due to the broad geographic distribution of the hubs, this process involved careful consideration of logistics, which attempted to transfer ownership and control of the process and equipment to Northern colleagues. To manage remote collaboration, each hub assigned a team lead whose responsibilities included organizing a physical meeting space for workshops, receiving equipment mailed from southern Canada, and documenting the development process. We encouraged local 3D printing of components, which took place in two communities: Inuvik (at the Arts, Crafts & Technology Micro-manufacturing Centre; see Figure 3) and Yellowknife (at the Smart Communities Society NWT design space). However, as discussed below, these intentions faced several limitations in practice.

The next three sections present findings from interviews and discussions with the NWT innovation hubs, to highlight lessons learned with respect to codesigning community networking literacies in the context of rural/remote NWT communities.

Findings from intake interviews

The 18 intake interviews informed the first steps of our community networking literacy codesign process. First, participants stressed contexts of land-based activities, Indigenous knowledge protocols, and a history of settler colonialism as expressed in resource extraction, political and economic disempowerment, and residential schools. They noted climatic and geographical elements present in Northern winter contexts, such as limited daylight (24-h darkness) and cold temperatures (e.g., −50°C). One participant indicated that "[I] immediately wonder about its [CN] potential in this kind of [harsh] climate" (p. 16). Participants also identified capacities and assets inside NWT communities that could support this initiative, such as pre-existing 3D printing facilities in Inuvik and Yellowknife. They raised potential social and technical

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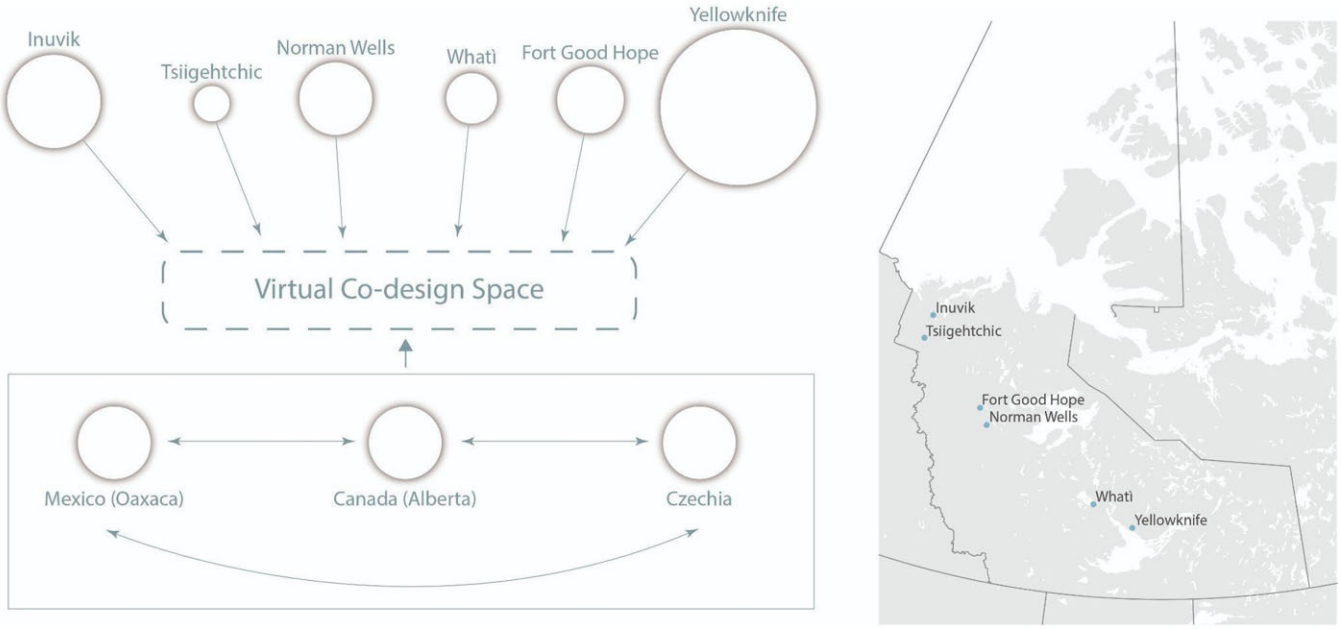


Figure 2. Graphic of NWT-based Innovation Hubs. The circle size represents population size, using the NWT government's population estimates as a guide: https://www.statsnwt.ca/population/population-estimates/bycommunity.php. Image Credit: Murat Akçayir.



Figure 3. Arts, Crafts & Technology Micro-manufacturing Centre (ATMC) in Inuvik, NT. Left photo credit: George Parkes; Right photo credit: Lars Ekelund.

barriers to the successful implementation of CNs, including cost and maintenance of equipment. One interviewee said: "You've got to find those people who are willing to [sustain the CN]" (p. 6). Participants noted that community members may not recognize a need for CNs or may not be optimistic about the idea, particularly in the context of distrust from years of extractive research, and the potential of bad actors to exploit CNs for profit, gossip, gambling, and copyright-infringing content. This highlighted a desire among participants for local control of the content and operations of CNs.

Second, participant contributions reflected an orientation toward sustainability and scalability. Emergent themes included a desire for the project to respond to limitations around Internet connectivity and affordability. This also involved paying close attention to social processes of capacity-sharing, relationship-building, and digital knowledge stewardship and exchange. Participants requested clarity on the ownership, governance, and administration of the CNs, and suggested developing local agreements regarding access and use. They highlighted the need to find appropriate collaborators and provided advice about community engagement with Elders and youth. They stressed the project should contribute to the empowerment of community members and ensure the transfer of tacit knowledge associated with the

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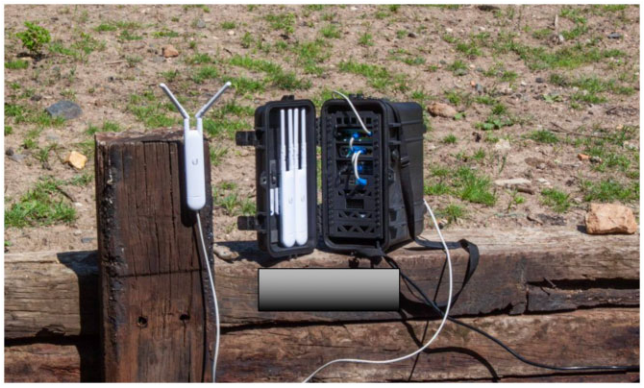


Figure 4. Image of Nimble Mesh Networking Kit Presented to Interview Participants. Photo credit: Eric Nitschke.

assembly and operations of the mesh networking kits. These discussions confirmed a strong interest in values such as self-determination and technological autonomy.

Participants stressed the codesign process should be undertaken in a manner that supports remote coordination due to geographic distances and cultural gaps between team members. For example, they recommended setting up teams of community-based codesigners, to find ways to share and build technical capacity. These suggestions inspired our creation of the NWT innovation hubs. While some participants expressed reticence about the technical nature of the project, most felt comfortable learning more. When presented with a photo of the mesh networking kits (see Figure 4) and asked how they felt about its physical design, people used words like "fascinated," "cool," and "wouldn't be too technical." Some community members suggested that Elders or others might be intimidated: "The older generation [may say] 'wow this looks too technical and too complicated'" (p. 15).

Third, we examined the cultural translation of technical and organizational concepts by testing metaphors and analogies about CNs. We asked participants how to ensure the project reflected ways of life in rural Northern Indigenous communities and how we might build shared understandings alongside people with no prior experience with mesh networking technologies. After presenting the draft analogy of a community freezer introducing the concept of CNs in relation to shared resources, ownership and control, as well as local content, we tested it with community participants.[2] Most felt the analogy would resonate with their families, friends, and neighbors: "They will definitely understand that concept, I think that would really help with the idea of [CNs]" (p. 9). However, several noted the analogy was context dependent—in some cases, community freezer projects failed and so might not represent a sustainable community-owned technology. This variability reveals the importance of understanding local contexts in sociotechnical design and speaks to the diversity among NWT communities. Some participants suggested other analogies that might resonate better, including community radio ("This [CN] to some extent reminds me of Community radio" (p. 16)) and the bright orange, high frequency "bush radios" that have used in the NWT since the 1970s ("CB radios … gives them kind of an idea that it is a more local [technology]" (p. 9)).

Finally, participants helped ensure the mesh networking kits and associated curriculum would be relevant and useful. Many adopted a critical stance, asking questions about why, how, and for what purposes communities are connected, as well as who benefits and what the consequences are. In line with previous research, many Indigenous participants placed particular importance on Indigenous cultures and languages (Li et al., 2021; Perley et al., 2016). We then discussed applications to showcase, including local storage of media files (e.g., video and music) and ways to use the CNs to share community information.

Findings from codesign workshops

The first codesign workshop, "Let's Build the Black Box," focused on assembling the mesh networking kits. NWT hubs received a box of materials and set-up instructions and collectively built their kits. This provided an example of reimagining hardware, and the hands-on nature of the event encouraged interest among participants. However, we also experienced logistical challenges preceding the workshop, since it took approximately 8 weeks to order components and mail them to the NWT hubs, as some distributors will not ship hardware components directly to rural/remote NWT communities. 3D printing logistics presented another challenge, as only Inuvik and Yellowknife had access to facilities, suggesting that future project budgets include funding to purchase local manufacturing equipment such as 3D printers and filament. Despite these minor issues, the workshop demonstrated the viability of the virtual workshop format and the possibility of team building in online settings (see Figure 5). Even though many participants had never met in person, they expressed comfort in sharing their hopes and aspirations for the project, as well as working through the hardware assembly. Prior online meetings and interviews may have helped establish this rapport.

The second workshop, "Customizing the Interface and Picking Applications," explored the platform user interface and introduced the open-source applications to be pre-loaded on the mesh networking kits. The workshop goal was to involve innovation hubs in customizing the interface and content/services. After updating participants on project progress (Quarry & Ramirez, 2009), we introduced a demonstration "landing page" accompanied with a presentation describing the use of interfaces in inclusive online learning environments led by two hub members from Aurora College. Content included: designing for diverse learning styles (e.g., linear and intuitive); including multimodal content; recognizing mobile design; and personalizing visual contexts for local users. This presentation engaged participants as instructors and sparked reflections about the importance of designing the user interface to fit local NWT contexts. Next, we invited regional hub participants to draw the interface, with plans to translate their drawings into digital form. The limitations of the virtual setting led us to revert to a predesigned interface and invite feedback from participants (which was minimal). Despite limited participation in this activity, the ensuing discussion stressed the need to document organizational and tacit knowledge about the codesign process and generated questions on managing access to the user interface and applications (e.g., role access and administration). We then demonstrated four offline, open-source applications and services: Jellyfin (media-streaming), WordPress (webpage design), Jitsi (video chat), and Unifi-Controller (network management). The workshop ended by inviting participants to consider how to best support learners navigating the mesh networking kits for the first time.

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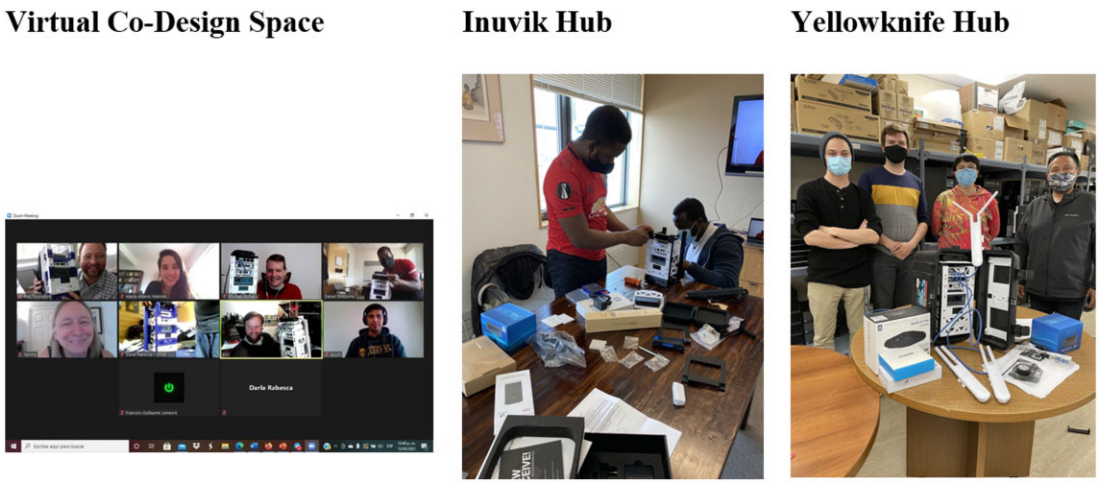


Figure 5. Images from Codesign Workshop 1: "Building the Black Box." Left photo credit: Maria Alvarez-Malvido. Centre photo credit: Ray Savard. Right photo credit: Amrik Kanwal.

The third workshop, titled "Installing and Configuring the Mesh Networking Kits," was originally designed to illustrate the remote installation of the firewall and operating system. However, given the technical nature of this activity, we adjusted the format to one-on-one meetings between the design team and the NWT hub leads. Over the course of approximately 30 paired support sessions, the design team remotely installed the operating system software, open source "Lokal" services platform, and other software.[3] This enabled the design team to test the syncing of content and backups between individual kits and global servers, and establish procedures for receiving remote support and updates. This generated helpful lessons regarding the sequencing and preparation of software, as well as suggestions for hardware adjustments, resulting in a more integrated and streamlined installation process (Taneja, 2022). Despite these benefits, this approach limited the participation and input of hub participants, and relied on centralized processes that pushed content out to local mesh networking kits. The design team discussed these limitations and decided that unless hubs held the technical expertise to perform these activities themselves, such an approach is necessary.

The fourth workshop, "Let's Test Out the Apps," began with a check-in that determined that not everyone had a working mesh networking kit, as many hubs were still installing the OS and software (as described above) or waiting for hardware. Therefore, the workshop focus shifted to a demonstration by the design team rather than a hands-on activity led by the hubs, with community members watching via synchronous Zoom. We demonstrated how the mesh network is composed of APs managed by real-time monitoring software and illustrated how mobile devices are passed between wireless APs. Participants asked whether communities could interconnect and how CNs might expand inside a community. In response, the team noted that the kits were designed for use inside a community, but that interconnections are possible with affordable longer-range antennas, and expansion within a community was simply a matter of adding additional APs.

The workshop also demonstrated the open source applications JellyFin media streaming and Jitsi chat. Participants asked whether each unit could have its own separate JellyFin media library, and whether local and Indigenous content from NWT media creators such as the Inuvialuit Communications Society can be included. Both independent JellyFin libraries and content from NWT creators could be enabled in the kits, although the latter would require a negotiation of permission for any copyright-protected content. They also raised questions related to backup power (included in the kits) and user privacy. This highlighted the importance placed on protecting community end-users. The workshop ended with a request to test the kits "in the field."

The fifth and final workshop, "Administration and Governance of Mesh Networking Kits," sought to generate reflections related to ownership and control of content and services, as well as privacy and ethics. At this point, 14 kits were in various states of assembly and operation, with seven units distributed to NWT hubs and the remainder with university researchers. During project check-ins, participants, many of whom had no formal technical training, reported on the status of technical components such as the OS and firewall configurations. The group discussed ongoing administration, ownership, and control of the kits. Recognizing project funding would expire on March 31, 2022, we presented a draft Stewardship agreement outlining roles and responsibilities, and the NWT hubs reviewed and signed the agreement. The workshop ended by discussing next steps, with participants expressing interest in exploring CN use cases, experimenting with open-source apps, and developing operational models (ownership and control of the local mesh network systems). Participants recommended a path forward: finalize building the mesh networking demonstration kits, present them to local leadership, and engage the community to support local deployment.

Post workshop interviews and reflections

Following the final workshop, a university researcher conducted one-on-one exit interviews with seven NWT hub members. Participants raised concerns about workshop timing, logistics, and content, as well as perceived strengths and limitations of the codesign process. They suggested how to improve future workshops and sustain the project. During the year-long project, participation had fluctuated, and enthusiasm for the project and availability to join workshops varied due to busy schedules and

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long gaps between events. Interviewees noted the challenges of connecting with their hub, leading to suggestions to draw upon existing communities of interest, such as among Aurora College community adult educators and local youth. Further, some people experienced technical difficulties during Zoom workshops and with connecting their mesh networking kits, pointing to barriers including limited Internet access and challenges with hardware set-up.

Overall, participants appreciated the workshops. One person stated, "I'm amazed that we made it work … [T]he fact that we actually pulled it off feels like a fairly big accomplishment" (p. 8). However, participants also noted limitations. They requested more regular contact, such as through technical support and debrief meetings. One person noted, "As we were kind of going through the workshops, it was also being designed. So, it kind of slowed down the process and we hit some bumps" (p. 26). Several people noted the slow progress. At times, long gaps between events dampened enthusiasm: "The regularity of meetings [was an issue] … I was so excited. I was like 'okay, let's get started', [and] then we add to it like another two months or so before we had another get together" (p. 7). One person's frustrations compounded because their kit was not working at the end of the project. As a result, hubs ended up with kits at different stages of assembly and use.

With respect to workshop content, some participants felt it was too technical and complicated. Others did not feel comfortable or confident explaining what the mesh networking kits are, or how they can be used. As one person said, "I'm getting ideas on what it [the kit] will do, but I'm still not comfortable talking to people about what it will do" (p. 20). Some were unsure how to fix things when they broke: "What happens when it fails?" (p. 17). Several stated they need more training on how to set up and maintain the units (p. 17). Some participants admitted they felt confused after the workshops ended.

> So, the workshop was great. It told us about the parts and stuff like that. How to put it together. But then we were kind of left with all these wires and we didn't really know where they went … I requested diagrams and people sent me diagrams … but I think until he [design team member] actually walked me through it, I didn't get it—and I'm pretty good with technology (p. 20).

Others experienced information overload, particularly if new to these topics:

> I just felt sometimes like I was a fish out of water. Whenever you're talking about something about computer RAM or something and I'm trying to process it as we go along … I'm trying to listen, and then, if I don't process it because there's no interaction, while they're talking, it's like, okay, now I'm lost … " (p. 23).

Although participants identified concerns with timing, gaps, and technical aspects of the workshops, they also identified many benefits. Several were pleasantly surprised at the accessibility of the technical information. "I was surprised I wasn't frustrated with the technical explanations. … I realized how easy the process is. It's not impossible, but [it] requires thinking about things differently" (p. 17). Another person reflected at length on the process:

> As far as co-design, I didn't feel that involved with that part of it. But you know, being able to participate by assembling things and connecting things in and being the, I don't know, kind of being the remote-control robot that did those things—that was, yeah, that was definitely co-design. And then, you know … simply being consulted and you know, being able to put in some ideas about what would work with people in the community, what they would expect. Being able to bring that in felt very much like a co-design process. (p. 8).

The process helped participants reimagine infrastructure, an important component of community networking literacy. They expressed that the mesh networking kits are helpful in explaining the Internet, and one noted that although local networks can be challenging to understand, making the design process visual and tangible helps. "I see such potential in this [project] … It would also help explain the Internet on a smaller scale, because I think people think, 'the Internet' and they get overwhelmed with what it is. Whereas the [mesh networking kit] is a smaller scale that just covers this community" (p. 20). Overall, participants expressed enjoyment when assembling the kitsand with the workshop demonstrations: "It's an open thing, and it's not hiding anything under a veneer of nice sheet metal or plastic … It seems useful that it somehow demystifies it and makes it something that anybody could actually learn to do" (p. 8). Other highlights included moving from conceptual maps (Make the NetWork) to building and deploying physical mesh networks.

Participants highlighted the social element of the remote codesign process: "Having you guys online and assembling the case together … It's those things that we did together that really, really shine for me" (p. 8). Some participants contacted each other in between workshops: "It's interesting to see … you know, talking to people and finding out how they're making out with it" (p. 20). Others noted that being 'forced' to physically interact with the network components increased their confidence and made the process of building a CN more tangible. Rather than having instructors come to build the CN for a community, participants noticed the remotely facilitated workshops encouraged engagement. One person stated, "I almost gives you ownership over it. It makes you feel proud … [and you think] it's going to be amazing once we're able to do something with it" (p. 20). Another reflected on this point in detail:

> If you guys had been able to come up here, you might have been a whole lot more hands on and we [would] just have sat back and let you do things. So, I think there's some advantages to working with virtual connections. I mean somebody's got to have their hand on the screwdrivers and the cables, but that it was, you know, our hands in the community, that was, I think, a big plus … it drove us to try things that if you'd have been here, we might have just sat back and let you do it … It's forcing us to build capacity here, because either that, or it just doesn't get done. (p. 8).

Participants suggested how to enhance future workshops, such as through more multimedia content. For example, a pre-recorded demonstration video could show what CNs are and what they can do in the context of rural/remote NWT communities. As one person said, "When we're thinking about 'non-computerized' people, they don't care about the parts and the wiring. All that they want to know is what it [CN] can do." (p. 20). Participants requested more hands-on

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opportunities to test the kits and suggested more opportunities for technical codesign, for example, when canvassing relevant software and open-source resources.

Participants provided ideas for future mesh networking kit use cases, for example, to manage local data, access services like online education, or provide safety while people are out on the land. Participants suggested sharing locally and culturally relevant content, such as videos about making caribou hide tipis and birch bark canoes, and having Elders telling stories: "[It could help] to keep the language alive, keep the traditional way of learning alive" (p. 20). Another linked the kits to old HF Bush radios and newer social media, " ... building from that desire to reach out and connect to somebody" (p. 8), such as through live-streaming community events and engaging in instant messaging/voice/video calls.

Finally, we asked for advice to encourage the sustainability of the project. Participants suggested engaging community leadership and involving multiple project stewards or custodians: "I love being a champion for this, but I'm not going to be around forever ... You've got me thinking about that sustainability—don't just depend on one champion" (p. 8). Another stated: "We have to have community members involved in it for sure, and not just myself who you know, maybe will retire, hopefully in four years" (p. 20). Another person pointed out the importance of building connections between participants, such as by posting personal biographies and photos or hosting online social events.

Conclusion

Our research generated important reflections on both the possibilities and the limitations of remotely codesigning community networking literacies with rural Northern Indigenous communities. Several tensions emerged through interviews and workshops, and while we remain excited about the project's potential, we also recognize the limited participation and representation of Northern and Indigenous voices, logistical and process challenges, and gaps in technical and educational processes. Below, we examine these themes and answer our research question: *In what ways can the social and technical aspects of community mesh networking be adapted by people in NWT communities to develop appropriate forms of infrastructural literacy?*

With respect to community engagement, we approached participants as interested individuals rather than community representatives; our engagement does not reflect any formal mandate from community leadership (whether political or cultural). Although the project leveraged existing connections, many participants stressed that the CNs would only be successful with greater buy-in from the community. Geographic, technical, and cultural factors further restricted the substantial participation of more Northern Indigenous community members in the codesign of educational resources. These conditions threaten to reproduce and widen existing inequalities experienced by already marginalized Northern residents. We heard reports of communications challenges within NWT hubs, and between hubs and the research team, which was reflected in uncertainty about the ongoing ownership and control of the mesh networking kits. Furthermore, one hub (Déline) did not proceed as planned because our key contact moved to a different community, which supports prior research pointing out the limited sustainability of projects tied to specific individuals and places (Lithgow et al., 2022). NWT participants stressed the importance of including land-based activities, Indigenous knowledge protocols, and a recognition of settler colonialism and extractivism in educational material.

Regarding tensions between the coordination and pace of work, several participants found the delays between workshops diminished momentum, while others felt overwhelmed with technical minutiae. Working across multiple NWT hubs also reveals the limits to which community networking literacy educational projects can use standardized resources and codesign processes. Our team attempted to develop educational resources in a reflexive and emergent manner that incorporated participant feedback; however, this was complicated by engaging in virtual codesign over a large geography and under the restrictions of a global pandemic. This also highlights how codesign under remote conditions must balance inclusion and trust with enthusiasm and engagement. Educators looking to codesign CN learning materials must carefully consider how tacit knowledge developed in one community can best be transformed into generalizable and explicit insights.

Participants raised social and technical barriers to the sustainability of CNs in their communities, particularly in the context of distrust from years of extractive research, and the potential of bad actors to exploit connectivity for antisocial activities. Strong emergent themes included a desire for course material to respond to community needs regarding limited connectivity and affordability challenges, as well as support local control of connectivity and digital applications. This involves attending to social processes of capacity-sharing and knowledge stewardship, and underscores the need for appropriate community collaborators, including Elder and youth representatives. Participants also noted that the project needed clarity on the ownership, governance, and administration of hardware, and suggested developing local agreements regarding access and use. Finally, they identified useful capacities and assets inside NWT communities, including 3D printing facilities in Inuvik and Yellowknife.

Along with values, our experience highlighted the importance of shaping educational resources to fit varying levels of technical and cultural knowledge—on the part of both community participants and the research/design team. While NWT hub participants expressed concerns about the technical nature of course material, they also educated the UAlberta-based research team on the cultural and social elements of this work. Discussions related to culturally appropriate metaphors for mesh networks (whether community freezers, radio stations, or HF "bush" radios), ongoing debates around the ownership and control of local technical systems, and ideas for the potential uses of CNs reflect examples of these issues. NWT hubs continually reminded researchers to critically reflect on why, how, and for what purposes communities are connected, as well as on consequences.

These observations are particularly important in the case of community networking literacies, where the immediate use cases *may* be recognizable to communities, but the path to deployment and long-term sustainability is vague. Interviews stressed the importance of considering end use applications that spark interest among community members, which ranged from addressing gaps in local connectivity and high data overage fees, to building alternative community-controlled connectivity systems that support community-based social, cultural, and economic activities.

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Many participants identified the hardware assembly workshop as a highlight, describing it as a powerful, hands-on example of demystifying hardware. This suggests that community networking literacy initiatives should cover the material aspects of network design and deployment. While remote pedagogy requires hardware and software configuration to be completed in advance by the research/design team, participants benefit from a "hands on" understanding of the hardware and software.

Collectively, none of these findings suggest that the remote codesign of community networking literacies via distributed virtual innovation hubs is impossible. However, our experiences reveal that successful projects are fundamentally local in nature. Even for the smallest communities (four of the NWT hubs were based in communities of less than 700 people), literacies often take local forms. Metaphors, use cases, and local champions all differed among communities, illustrating there is no single optimal path for codesigning educational resources. While the community freezer analogy may resonate in one community, it does not apply universally. Similarly, educational use cases for the mesh networking kits were popular but were by no means the only potential use. In a sharp rejection of the "one size fits all" approach of curriculum design, our experience illustrates that community-led networking literacies must be localized and approached on community terms. This finding is our primary contribution to digital literacy research and practice.

Moving forward, we hope Northerners remain interested in the testing, deployment, and evaluation of community networking educational resources. We envision that NWT participants will work with the research/design team to test their kits in local geographic and community contexts—showcasing the mobility and services made possible through local wireless mesh network systems. Ideally, these activities will build toward participants presenting to local leadership and community members, with the goal of building sustainable CNs that reflect and support their desire for self-determination and technological autonomy.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author, R.M. The data are not publicly available due to their containing information that could compromise the privacy of research participants.

Funding

This work was supported through the DigitalNWT project, which received funding through a Digital Literacy Exchange Program grant from Innovation, Science and Economic Development Canada [grant number: 511811].

Conflict of interest statement

The authors declare that there is no conflict of interest.

Notes

1. To download these OERs visit: www.digitalNWT.ca/courses
2. To view the image of the communal freezer in Tuktoyaktuk used during these presentations, visit: http://topoftheworldgirl.weebly.com/home/tuktoyaktuk-ice-house (top-left image).
3. See: https://github.com/Wakoma/Lokal

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